Chapter 2 Chapter Test

Proportions, Similarity, Measurement, and Percent

Directions: Solve each problem. Show all work for full credit. Circle or box your final answer.

1. Solve the proportion.

$$\frac{k}{8} = \frac{-5}{2}$$

2. Solve the proportion.

$$\frac{5}{y} = \frac{7}{4} - \frac{3}{y}$$

3. Solve the proportion.

$$\frac{7}{w+4} = \frac{-7}{w}$$

4. Solve the proportion.

$$\frac{3}{8} = \frac{k-17}{32}$$

5. Solve the proportion.

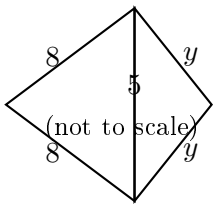
$$\frac{y}{8} = \frac{4}{15}$$

6. Solve the proportion.

$$\frac{n-3}{4} = \frac{n+1}{6}$$

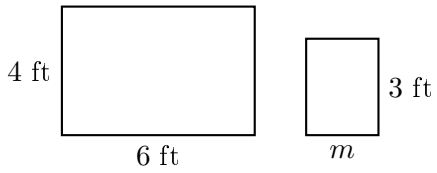
7. A boat travels 125 miles in 10 hours. Write and solve a proportion to find the speed (in mph) the boat must travel to cover the same distance in 8 hours.

8. The two figures below are similar. Find the value of y.



Side lengths: left figure has sides $8, 8, 5$*; right figure has sides* $y, y, ?$*.*

9. The two rectangles are similar. Find m.



10. A 3.5-ft child casts a 60-inch shadow. At the same time, a nearby pole casts a shadow that is 50 ft long. How tall is the pole? Write and solve a proportion.

11. Find the percent of change. State whether it is an *increase* or *decrease*.

Original: \$5,000 New: \$6,500

12. Find the percent of change.

Original: 150 lb New: 135 lb

13. A circular running track has a circumference of 100 ft.

(a) Find the diameter of the track. (Use $\pi \approx 3.14$.)

(b) Find the area of the region enclosed by the track.

14. A map uses the scale $\frac{1}{2}$ in $= 25$ miles. Two cities are $3\frac{1}{4}$ inches apart on the map. What is the actual distance?

15. A scientist surveys 30 flowers and finds 6 hummingbirds visiting them. If there are 525 flowers in the field, predict how many hummingbirds are visiting.

16. A length is measured as $\frac{3}{8}$ inch to the nearest $\frac{1}{8}$ inch, OR as 10 mm to the nearest millimeter. Which measurement is *more precise*? Explain.

17. A 10-pack of CDs costs \$3.17. A 50-pack costs \$13.00. Which is the better buy? Show your unit-price calculation.

18. A student claims:

> *"A computer originally priced at \$1,000 is now \$1,500. The percent of change is* $\frac{1500}{1000} = 150\%$*."*

Identify and correct the student's error.

Chapter 2 Chapter Test – Detailed Answer Key

1. Solve $\dfrac{k}{8} = \dfrac{-5}{2}$.

Cross-multiply:

$$2k = 8 \times (-5) = -40 \implies k = -20.$$

Answer: $k = -20$

2. Solve $\dfrac{5}{y} = \dfrac{7}{4} - \dfrac{3}{y}$.

Move the $\dfrac{3}{y}$ term to the left:

$$\frac{5}{y} + \frac{3}{y} = \frac{7}{4} \implies \frac{8}{y} = \frac{7}{4} \implies 7y = 32 \implies y = \frac{32}{7} \approx 4.57.$$

Note: some editions write the proportion so that cross-multiplication gives $y = \frac{22}{5} = 4.4$. *Check the original equation carefully. With the form above,* $y = \frac{32}{7}$.

Answer: $y = \dfrac{32}{7} \approx 4.57$

3. Solve $\dfrac{7}{w+4} = \dfrac{-7}{w}$.

Cross-multiply:

$$7w = -7(w+4) \implies 7w = -7w - 28 \implies 14w = -28 \implies w = -2.$$

Check: $LHS = \dfrac{7}{-2+4} = \dfrac{7}{2}$; $\quad RHS = \dfrac{-7}{-2} = \dfrac{7}{2}$. ✓

Answer: $w = -2$

4. Solve $\dfrac{3}{8} = \dfrac{k-17}{32}$.

Cross-multiply:

$$3 \times 32 = 8(k-17) \implies 96 = 8k - 136 \implies 8k = 232 \implies k = 29.$$

Answer: $k = 29$

5. Solve $\dfrac{y}{8} = \dfrac{4}{15}$.

Cross-multiply:

$$15y = 32 \implies y = \frac{32}{15} \approx 2.13.$$

Answer: $y = \dfrac{32}{15} \approx 2.13$

6. Solve $\dfrac{n-3}{4} = \dfrac{n+1}{6}$.

Cross-multiply:

$$6(n-3) = 4(n+1) \implies 6n - 18 = 4n + 4 \implies 2n = 22 \implies n = 11.$$

Check: $LHS = \dfrac{8}{4} = 2$; $\quad RHS = \dfrac{12}{6} = 2$. ✓

Answer: $n = 11$

7. Boat speed proportion.

At the original rate: speed $= \dfrac{125}{10} = 12.5$ mph.

Set up the proportion:

$$\frac{125}{10} = \frac{125}{t} \implies t = 10 \text{ hours (same distance)}.$$

To cover 125 miles in 8 hours:

$$r = \frac{125}{8} = 15.625 \text{ mph} \approx 15.6 \text{ mph}.$$

Answer: 15.625 **mph** $\approx$ 15.6 **mph**

8. Similar bowtie figures – find y.

The two triangles share a vertex and are similar. The corresponding sides are in proportion. The left triangle has equal legs of length 8 and base 5; the right triangle has equal legs of length y and the same shared base 5.

$$\frac{8}{y} = \frac{8}{5} \implies \text{(using the base-to-leg ratio)} \quad \frac{5}{8} = \frac{y}{5} \implies y = \frac{25}{8} = 3.125.$$

Alternatively, if the corresponding sides give $\dfrac{8}{5} = \dfrac{5}{y}$*:*

$$y = \frac{5 \times 5}{8} = \frac{25}{8} = 3.125 \text{ cm}.$$

Answer: $y = 3.125$ **(cm, or whatever units given)**

9. Similar rectangles – find m.

The rectangles are similar. Match corresponding sides:

$$\frac{6}{m} = \frac{4}{3} \implies 4m = 18 \implies m = 4.5 \text{ ft}.$$

Check: ratio of widths $= \dfrac{6}{4.5} = \dfrac{4}{3} =$ *ratio of heights.* ✓

Answer: $m = 4.5$ **ft**

10. Shadow / similar triangles – height of pole.

Convert child's shadow to feet: $60 \text{ in} = 5 \text{ ft}$.

Set up the proportion (height to shadow):

$$\frac{3.5}{5} = \frac{h}{50} \implies h = \frac{3.5 \times 50}{5} = \frac{175}{5} = 35 \text{ ft}.$$

Answer: The pole is 35 ft tall.

11. Percent of change: \$5,000 to \$6,500.

$$\%\text{ change} = \frac{6500 - 5000}{5000} \times 100 = \frac{1500}{5000} \times 100 = 30\%.$$

Since the value increased, this is a **30% increase**.

12. Percent of change: 150 lb to 135 lb.

$$\%\text{ change} = \frac{135 - 150}{150} \times 100 = \frac{-15}{150} \times 100 = -10\%.$$

This is a **10% decrease**.

13. Circular track: $C = 100$ ft.

(a) Diameter:

$$C = \pi d \implies d = \frac{C}{\pi} = \frac{100}{3.14} \approx 31.85 \text{ ft}.$$

(b) Area:

$$r = \frac{d}{2} \approx 15.924 \text{ ft}, \qquad A = \pi r^2 \approx 3.14 \times (15.924)^2 \approx 3.14 \times 253.57 \approx 796.2 \text{ ft}^2.$$

Answers: $d \approx 31.85$ ft; $A \approx 796$ ft^2

14. Map scale: $\frac{1}{2}$ in $= 25$ mi; distance $= 3\frac{1}{4}$ in.

$$\frac{0.5 \text{ in}}{25 \text{ mi}} = \frac{3.25 \text{ in}}{d} \implies d = \frac{3.25 \times 25}{0.5} = \frac{81.25}{0.5} = 162.5 \text{ mi}.$$

Answer: 162.5 **miles**

15. Hummingbirds: 6 per 30 flowers; predict for 525 flowers.

$$\frac{6}{30} = \frac{h}{525} \implies h = \frac{6 \times 525}{30} = \frac{3150}{30} = 105.$$

Answer: 105 hummingbirds

16. Precision: $\frac{3}{8}$ inch vs. 10 mm.

The precision of a measurement is determined by the size of the smallest unit used.

- $\frac{3}{8}$ inch to the nearest $\frac{1}{8}$ inch: $\frac{1}{8}$ in $= 3.175$ mm.
- 10 mm to the nearest 1 mm: unit $= 1$ mm.

Since 1 mm $<$ 3.175 mm, the millimeter measurement uses a **smaller unit**, so it is **more precise**.

Answer: 10 mm to the nearest millimeter is more precise.

17. Better buy: 10-pack at \$3.17 vs. 50-pack at \$13.00.

Unit price (cost per CD):

$$\text{10-pack}: \quad \frac{\$3.17}{10} = \$0.317 \text{ per CD.}$$

$$\text{50-pack}: \quad \frac{\$13.00}{50} = \$0.26 \text{ per CD.}$$

The 50-pack costs less per CD.

Answer: The 50-pack is the better buy (\$0.26/CD vs. \$0.317/CD).

18. Identify and correct the student's error.

Student's work (incorrect):

$$\frac{1500}{1000} = 1.5 = 150\%.$$

Error: The student divided the *new price* by the *original price*. The percent of change formula requires dividing the *change* (new − original) by the *original*.

Corrected solution:

$$\%\text{ change} = \frac{\text{New} - \text{Original}}{\text{Original}} \times 100 = \frac{1500 - 1000}{1000} \times 100 = \frac{500}{1000} \times 100 = 50\%.$$

The computer price increased by **50%**, not 150%.

End of Chapter 2 Chapter Test Answer Key